

FDD3024: Module 3 Oral Exam Problems

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In these oral exam problems, we consider the definition of the IMP language and its semantics and Hoare rules from the lecture 10 slides.

1. Soundness of Hoare rules for IMP

- a) Precisely define semantic validity for Hoare triples in terms of the operational semantics of IMP.
- b) Formulate the soundness theorem for the IMP Hoare rules and sketch a proof that justifies each given rule.

2. GCD Program in IMP

Consider again the GCD function from the module 1 homework:

```
let rec gcd (a : int) (b : int) : int =  
  if a = 0 then 0  
  else if b = 0 then 0  
  else if a = b then a  
  else if a < b then gcd a (b-a)  
  else gcd (a-b) a
```

Write an IMP program that implements this function using a single while loop. Use the given function as a *functional model* in a contract (pre- and postcondition) for the IMP program. Annotate the while loop with an invariant. You need to argue informally why your invariant is correct and ensures the postcondition.